

Ron Isaac Otieno

Nairobi, Kenya | isaacron195@gmail.com | github.com/isaac-ron | linkedin.com/in/ron-otieno | [Portfolio](#)

Computer science student (graduating December 2026) building full-stack products, machine learning systems and data pipelines. Seeking software engineering internships or junior roles, and data analyst or data engineering roles.

EXPERIENCE

Freelance Web Developer

Independent

Nairobi, Kenya

Apr 2026 – Present

- **Association of Care Leavers Networks in Africa (ACNA):** for the launch of its flagship East Africa chapter, built the network's membership application and the websites of two founding member networks, Virtue Literacy Africa and Second Chances Kenya. Staff manage content through a git-based headless CMS (Astro, Keystatic).
- **The Nile Explorer:** rebuilt a regional media network's news site on Next.js and Sanity, with on-demand revalidation on publish and readership-ranked Top stories; migrated the WordPress archive with a repeatable, dry-run-capable script and wrote a handover manual for the newsroom.
- Gathered requirements directly with clients and delivered on self-managed timelines.

IT Support Intern

Kenya Revenue Authority, IT Service Delivery Department

Nairobi, Kenya

Jan 2026 – Mar 2026

- Provided first-line technical support and troubleshooting for enterprise hardware, software and network issues, keeping downtime on critical tax administration systems to a minimum.
- Documented user issues, tracked resolutions and communicated status updates across teams, coordinating with senior IT staff to implement fixes.

PROJECTS

FeeDesk – Multi-tenant School Fee Platform | React, Node.js, MongoDB, M-Pesa Daraja, Bank IPN APIs

Jan 2026 – Present

- Designed and built the backend services and database schema for a multi-tenant platform that reconciles M-Pesa and bank payments against student fee balances.
- Integrated Safaricom Daraja webhooks and bank IPN APIs as a passive payment listener feeding one reporting view.
- Led product scoping and requirements gathering, including data-compliance research. Running on staging ahead of a planned school pilot.

CrisisConnect – NLP Crisis Detection & Triage | TypeScript, Node.js, PostgreSQL, Python, ONNX

Sep 2025

- Built the backend and data pipeline for a crisis-reporting app, from raw report ingestion (including offline-queued reports) through model classification to a real-time alert API.
- Fine-tuned two RoBERTa classifiers evaluated on held-out disasters: priority macro F1 0.416 (baseline 0.273) and urgent recall 60% (baseline 36%); int8-quantized to ONNX (515 MB to 130 MB) to run on a free 512 MB instance.

WasteNet – CNN Waste Classification | Python, Flask, TensorFlow/Keras, Raspberry Pi 4, Arduino Uno

Sep 2025

- Fine-tuned a CNN for recyclable/organic sorting (88.9% accuracy, 94.2% precision, 92.7% recall), served from a Flask backend on a Raspberry Pi 4 driving Arduino sorting hardware, with a dashboard for live and uploaded-image inference.

EDUCATION & CERTIFICATIONS

Kabarak University

BSc Computer Science – on track for First Class Honours (CGPA 74%), coursework complete

Nakuru, Kenya

Expected Dec 2026

Coursework: Data Structures & Algorithms, Database Systems, OOP, Web Development, Machine Learning

ALX Data Engineering Program

May 2026 – Present

- Building ETL pipelines in Python, orchestrated with Airflow, containerised with Docker and scaled out with Spark.
- Covering batch and streaming ingestion, data modelling and architecture, and data security practices.

ALX Data Analytics Program with Professional Skills

Sep 2025 – Apr 2026

- Project-based program in data cleaning, exploratory analysis and visualisation, combining multi-source datasets into dashboards with SQL, Excel, Power BI and Pandas.
- Professional skills: time management, prioritisation, communication and cross-functional teamwork.

LEADERSHIP & ACTIVITIES

Vijana Empowerment Initiative CBO – Volunteer Web Developer & Trainer

May 2026 – Present

- Built the website for a Bomet youth training organisation; assist with web development training for local youth.

Egerton JavaScript Conference Hackathon – Winner

Mar 2025

- Led a team of three to build a customer-service platform (React, MongoDB) with OCR-based QR check-in using Tesseract.js, cutting average response time by 25%.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

Web & Backend: React, Next.js, Node.js, Express, Flask, Astro, REST APIs, M-Pesa Daraja API

Data & ML: PostgreSQL, MongoDB, Supabase, Pandas, Power BI, Excel, Airflow, Spark, TensorFlow/Keras, scikit-learn, Hugging Face

Tools & Cloud: Git/GitHub, Docker, AWS, Linux, Sanity, Keystatic